

Topological Stiffness and Geometric Characterization of a 36-Node Kuramoto Ring

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Abstract

We present a numerical characterization of a 36-node Kuramoto ring with nearest-neighbor coupling and Gaussian frequency heterogeneity. Three distinct dynamical regimes are identified: incoherent ($K < 3\sigma_\omega$), chimera ($3\sigma_\omega \leq K \leq 5\sigma_\omega$), and synchronized ($K > 5\sigma_\omega$). The chimera regime is characterized by the coexistence of coherent and incoherent oscillator clusters and corresponds to a maximum of geometric heterogeneity as measured by the standard deviation of the discrete Ollivier-Ricci curvature. Spectral analysis of the weighted graph Laplacian shows that algebraic connectivity (Fiedler value λ_2) saturates near $K \approx 4.5\sigma_\omega$, while Dirichlet energy and Ricci curvature variance approach their asymptotic minima near $K \approx 20\text{--}22\sigma_\omega$. These three independent geometric measures tell a consistent story: the ring transitions from a geometrically fragmented state to a topologically stiff, uniformly curved manifold. Parallels to biological coupled-oscillator systems (suprachiasmatic nucleus, cardiac SA node, cortical gamma oscillations) are noted.

1. Introduction

The Kuramoto model of coupled phase oscillators has been extensively studied for all-to-all coupling topologies. Nearest-neighbor ring topologies introduce qualitatively different behavior, including chimera states — spatially coexistent coherent and incoherent clusters — first described by Kuramoto and Battogtokh (2002) and subsequently observed experimentally in chemical, optical, and neural systems.

The present work characterizes the $N=36$ ring systematically from three geometric perspectives: (1) order-parameter dynamics and chimera detection, (2) discrete Ollivier-Ricci curvature on the graph, and (3) spectral properties of the weighted graph Laplacian. The combination of these three frameworks provides a complete geometric picture of the synchronization transition.

2. Model

The system is defined by:

$$\frac{d\theta_i}{dt} = \omega_i + \frac{K}{2} [\sin(\theta_{i+1} - \theta_i) + \sin(\theta_{i-1} - \theta_i)], \quad i = 1, \dots, N$$

with $N = 36$, periodic boundary conditions ($\theta_{N+1} \equiv \theta_1$), and natural frequencies ω_i drawn from a Gaussian distribution with zero mean and unit variance ($\sigma_\omega = 1$, normalized units). Coupling K is swept from 0.5 to $22 \sigma_\omega$.

The global order parameter is:

$$r(t) = \left| \frac{1}{N} \sum_{k=1}^N e^{i\theta_k(t)} \right|$$

3. Results

3.1 Three Dynamical Regimes

Numerical integration (RK45, max step $0.05 \sigma_\omega^{-1}$, integration time $80 \sigma_\omega^{-1}$) identifies three regimes by the stationary value of r :

Regime	K range	r (stationary)	Character
Incoherent	$K < 3\sigma_\omega$	$r \approx 0.1\text{--}0.2$	Random phases
Chimera	$3\text{--}5 \sigma_\omega$	$0.2 < r < 0.7$	Bistable, history-dependent
Synchronous	$K > 5\sigma_\omega$	$r \rightarrow 1$	Uniform phase locking

The chimera regime exhibits hysteresis ($\Delta r \approx 0.31$): starting from near-synchronous initial conditions yields higher r than starting from random initial conditions at the same K . This bistability is a hallmark of the chimera state in finite rings.

3.2 Ollivier-Ricci Curvature

The discrete Ollivier-Ricci curvature of edge $(i, i+1)$ is computed via optimal transport of probability measures supported on the 3-point neighborhoods of each node ($\alpha = 0.5$). Key findings:

- **Incoherent regime:** High curvature variance, ~40% of edges with $\kappa < 0$ (saddle-like). Geometrically fragmented.

- **Chimera regime:** Maximum curvature heterogeneity at $K \approx 3.69 \sigma_\omega$. Coherent clusters show $\kappa > 0$ (sphere-like); incoherent edges show $\kappa < 0$. This geometric coexistence is the Ricci signature of the chimera state.
- **Synchronous regime:** $\sigma(\kappa)$ decreases monotonically. At $K = 20\sigma_\omega$, variance is reduced by ~67% relative to the chimera peak.

3.3 Spectral Analysis of the Weighted Graph Laplacian

Edge weights are defined as $w_{\{i,i+1\}} = (K/2)|\cos(\theta_{\{i+1\}} - \theta_i)|$, the linearized coupling strength at the stationary state. The weighted Laplacian $L = D - W$ yields:

Algebraic connectivity (Fiedler value λ_2):

λ_2 grows monotonically with K but shows an inflection point near $K \approx 4.5 \sigma_\omega$. Beyond this, the graph is already a strong expander; additional coupling yields diminishing returns in connectivity.

Dirichlet energy:

$E = \frac{1}{2} \sum w_{\{ij\}}(\phi_i - \phi_j)^2$ measures geometric resistance to information transfer. E decreases monotonically, approaching its asymptotic minimum near $K \approx 18-22 \sigma_\omega$.

3.4 Consistent Geometric Picture

All three measures converge on the same narrative:

- $K < 3\sigma_\omega$: geometrically fragmented, low connectivity, high resistance
- $K \approx 3-5\sigma_\omega$: maximum geometric tension (chimera regime)
- $K > 5\sigma_\omega$: progressive geometric unification
- $K \approx 20-22\sigma_\omega$: near-asymptotic topological stiffness

3.5 Biological Analogies

The universal synchronization curve $r(K/\sigma_\omega)$ places three biological systems in context:

System	K/σ_ω (estimated)	Regime	Function
SCN (circadian)	~1.9	Near-incoherent	Flexible entrainment
SA node (cardiac)	~3.5	Chimera boundary	Robust pacemaking
Cortical gamma	~3.2	Chimera	Information processing

These are analogies based on literature parameter estimates, not derivations.

4. Discussion

The geometric characterization of the $N=36$ Kuramoto ring reveals that the chimera state is not merely a dynamical curiosity but has a precise geometric signature: maximum Ricci curvature heterogeneity, low algebraic connectivity, and high Dirichlet energy. The synchronized state corresponds to a topologically stiff, low-resistance, uniformly curved manifold.

The spectral saturation near $K \approx 4.5\sigma_\omega$ and the Dirichlet minimum near $K \approx 20\sigma_\omega$ define two characteristic scales of the system. The first marks the transition to strong-expander behavior; the second marks full geometric unification.

These findings are consistent across all three geometric frameworks, suggesting that the chimera transition in nearest-neighbor rings has a robust geometric interpretation that warrants further analytical investigation.

5. Conclusions

1. The $N=36$ nearest-neighbor Kuramoto ring exhibits three dynamical regimes with distinct geometric signatures.
 2. The chimera state corresponds to maximum Ollivier-Ricci curvature heterogeneity ($K \approx 3.69\sigma_\omega$).
 3. Algebraic connectivity saturates near $K \approx 4.5\sigma_\omega$; full geometric stiffness requires $K \approx 20\text{--}22\sigma_\omega$.
 4. Three independent geometric measures (Ricci variance, Fiedler value, Dirichlet energy) give a consistent picture.
 5. Parallels to biological oscillator networks are noted as contextual analogies.
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Code: Numerische Analysen in Python (NumPy, SciPy, Matplotlib). Verfügbar auf Anfrage.